

PRAYCG Master Comprehensive Suite v1.4.0

Detailed Analysis Module Reference

Core boundary: This reference explains what each module computes and how to interpret it. It does not claim that PRAYC-G proves meaning, OSM, hidden-Y biology, cellular photonics, or human EEG mechanism.

How to read this reference

The Master Comprehensive Suite is a layered post-processing system. Each module provides a constrained piece of evidence, not a standalone verdict.

The strongest interpretations require convergence: clean timing, clean signals, appropriate controls, self-report consistency, artifact veto, and model comparison.

CandidateLocal_KHT is new in v1.4.0. It converts exploratory event-level K_HT handoff analysis into a formal suite module while preserving the rule that physiology-discovered anchors remain exploratory in the same run.

Module	Primary output	Primary use	Boundary
stream_inventory	stream_inventory.csv	Confirms whether the raw ingredients for later modules exist.	A stream being present does not mean the signal is clean or interpretable.
marker_validation	events_used.csv, segments_used.csv	Prevents analyses from using the wrong time interval.	Software markers are intended timing unless validated by ALS/photodiode timing.
channel_map_validation	channel_map_used.csv	Determines which channels can be used for frontal, temporal, visual, and artifact-sensitive summaries.	A locked channel map is not source localization; it is a scalp-channel mapping aid.
cue_schedule_validation	cue_schedule_validation_summary.csv, cue_marker_events.csv	Protects the contextual-override task from accidental schedule mismatch.	Correct cue schedule does not guarantee task compliance.
override_task_scoring	contextual_override_task_scoring.csv	Provides behavioral evidence of analytic-task engagement.	An incorrect sum does not automatically invalidate Override; it indicates imperfect performance.
self_report_summary	self_report_ratings_long.csv, self_report_ratings_wide.csv	Keeps first-person report as an independent evidence stream.	Self-report does not prove physiology; physiology does not replace experience.
report_input_artifact_handling	artifact_exclusion_windows.csv	Prevents typing, movement, and report-entry artifacts from contaminating physiological windows.	The exclusion rule removes likely artifacts; it does not detect every artifact.

eeg_qc	eeg_channel_qc.csv when raw EEG is available	Determines whether EEG-derived modules are trustworthy enough to run.	Consumer EEG gamma is artifact-sensitive; QC has veto power.
mapped_lower_gamma	artifact_matched_condition_contrasts.csv	Tests whether Target, Override, and Control differ in gamma/work-like activity.	Lower gamma is a work signal, not direct proof of meaning.
residualized_lower_gamma	eeg_low_gamma_residualized_contrasts.csv	Separates candidate neural work from common-drive physiology when possible.	Residualization is only as good as the available covariates.
gammascalpel	gammascalpel_window_scores.csv, gammascalpel_score_contrasts.csv	Aims to separate analytic task-lock from candidate meaning-related work.	It is a scalp-level heuristic, not an anatomical or cellular measurement.
task_lock_plv	phase-lock or contrast tables when available	Helps distinguish forced analytic order from receptive narrative state.	High order is not automatically meaning; it may be task rigidity.
pac_exploratory	PAC tables when enabled	Looks for coupling between slow integration and faster work signals.	PAC is highly artifact-sensitive and should remain exploratory unless preregistered.
autonomic_api	autonomic_api tables when available	Adds body-state validation to EEG findings.	Cardiac response can reflect arousal, breathing, posture, threat, or emotion.
respiration_summary	respiration_summary.csv	Controls for respiratory common drive and helps interpret HRV/theta relationships.	Breathing can drive both heart and EEG features; it must not be ignored.
formal_pncc_theta	pncc_theta_artifact_matched_surrogate_contrasts.csv	Tests whether theta integration persists after a branch ends.	Video-end washout may miss meaning that resolves before the end of the clip.
tau_coh_theta	tau_coh_theta_summary.csv	Quantifies decay/persistence rather than only peak amplitude.	Requires sufficient clean post-event data and stable timing.
surrogate_theta_carryover	surrogate PNCC tables	Protects against seeing ordinary fluctuations as narrative afterglow.	Surrogates do not replace external replication.
post_meaning_cascade	frequency cascade tables	Looks for gamma/theta/alpha ordering consistent with intake -> integration -> settling.	A cascade pattern is not mechanism proof.
temporal_semantic_proxy	temporal_semantic_proxy_window_scores.csv	Creates a scalp-level proxy for semantic/narrative processing.	It is not STS/MTG source localization.
postpeak_pncc_theta	postpeak_peak_and_taper_detection.csv,	Implements the idea that integration follows the	Peak discovery is exploratory unless the rule

	postpeak_pncc_theta_contrasts.csv	meaningful intake peak, not necessarily the clip boundary.	is frozen before analysis.
annotation_locked_theta_carryover	annotation_locked_theta_carryover_contrasts.csv	Lets the analysis test scene/event anchors directly.	Retrospective subject clock-time estimates are weak anchors; media-structural anchors are stronger.
gamma_to_theta_handoff	gamma_to_theta_handoff_lag_correlations.csv, gamma_to_theta_handoff_best_lags.csv	Models gamma intake followed by theta integration.	Lagged correlation is not causation; random-anchor controls are still needed.
temporal_semantic_proxy_to_theta_handoff	gamma_to_theta_handoff_best_lags.csv	Tests whether semantic-proxy activity precedes theta carryover.	TSP is a proxy; it cannot prove semantic anatomy.
thresholded_state_update	thresholded_state_update_candidate_windows.csv, thresholded_state_update_summary.csv	Sharpens the scalpel when meaning occurs inside the clip rather than at the end.	Physiology-discovered windows remain exploratory in the same run.
state_locked_handoff_alignment	state_locked_anchor_summary.csv, state_locked_epochs_long.csv, state_locked_model_comparison.csv	Reduces temporal smearing across subjects or runs.	State-locking must be preregistered before it becomes confirmatory.
human_translation_kht	human_translation_kht_feature_frame.csv, phasewise_summary, cv tables	Tests rough reciprocal loop structure at the human PR-AYC-G level.	K_HT is not K_OSM and does not prove OSM.
candidate_local_kht	candidate_local_kht_analysis.csv, anchor_manifest, random_reference, interpretation.json	Formalizes the Her 113.9 s style exploratory handoff analysis inside the suite.	K alone is not enough; theta handoff alone is not enough; physiology-discovered anchors remain exploratory.
media_covariate_inventory	human_translation_kht_media_covariate_inventory.csv	Keeps raw sensory/input information separate from inferred biology.	Media covariates belong to $u(t)$, not $Y(t)$.
constrained_dtw_exploratory	DTW readiness notes / exploratory tables	Can compare shapes of state transitions despite speed differences.	DTW can overfit; use only as exploratory until locked.
stimulus_style_windows	configured windows in reports/tables	Makes the suite sensitive to different narrative structures.	Style presets are analysis assumptions, not facts.
master_endpoint_table	master_endpoint_status_table.csv	Provides a single executive table for interpretation.	A summary table cannot rescue weak source modules.
figures	figures folder	Supports review and	Figures are explanatory, not

communication.

proof.

stream_inventory

Lists the LSL/XDF streams present in the recording: EEG, markers, respiration, cardiac, ALS/AUX, and any auxiliary streams.

Primary outputs	stream_inventory.csv
Why it matters	Confirms whether the raw ingredients for later modules exist.
Interpretive boundary	A stream being present does not mean the signal is clean or interpretable.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

marker_validation

Checks event timing sources from XDF marker streams and local event logs. It builds the condition windows used for slicing Control, Target, Override, baselines, washouts, and report-input exclusions.

Primary outputs	events_used.csv, segments_used.csv
Why it matters	Prevents analyses from using the wrong time interval.
Interpretive boundary	Software markers are intended timing unless validated by ALS/photodiode timing.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

channel_map_validation

Loads the PRAYCG16 fixed channel map or a supplied map and records electrode/ROI labels and confirmation status.

Primary outputs	channel_map_used.csv
Why it matters	Determines which channels can be used for frontal, temporal, visual, and artifact-sensitive summaries.
Interpretive boundary	A locked channel map is not source localization; it is a scalp-channel mapping aid.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

cue_schedule_validation

Loads the cue schedule JSON/CSV, checks cue count and expected contextual-override sum, and verifies the task schedule used in the run.

Primary outputs	cue_schedule_validation_summary.csv, cue_marker_events.csv
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Why it matters	Protects the contextual-override task from accidental schedule mismatch.
Interpretive boundary	Correct cue schedule does not guarantee task compliance.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

override_task_scoring

Reads the subject's entered sum and compares it with the expected cue-schedule sum.

Primary outputs	contextual_override_task_scoring.csv
Why it matters	Provides behavioral evidence of analytic-task engagement.
Interpretive boundary	An incorrect sum does not automatically invalidate Override; it indicates imperfect performance.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

self_report_summary

Parses branch ratings for meaning, empathy, absorption, afterglow, analytic effort, discomfort, mind wandering, and task compliance.

Primary outputs	self_report_ratings_long.csv, self_report_ratings_wide.csv
Why it matters	Keeps first-person report as an independent evidence stream.
Interpretive boundary	Self-report does not prove physiology; physiology does not replace experience.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

report_input_artifact_handling

Creates exclusion windows around typed report/rating screens and other subject-input periods.

Primary outputs	artifact_exclusion_windows.csv
Why it matters	Prevents typing, movement, and report-entry artifacts from contaminating physiological windows.
Interpretive boundary	The exclusion rule removes likely artifacts; it does not detect every artifact.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

eeg_qc

Computes basic channel quality summaries such as amplitude, possible flatlines, extreme variability, or line-noise-sensitive behavior.

Primary outputs	eeg_channel_qc.csv when raw EEG is available
Why it matters	Determines whether EEG-derived modules are trustworthy enough to run.
Interpretive boundary	Consumer EEG gamma is artifact-sensitive; QC has veto power.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

mapped_lower_gamma

Computes lower-gamma condition contrasts in defined channel groups.

Primary outputs	artifact_matched_condition_contrasts.csv
Why it matters	Tests whether Target, Override, and Control differ in gamma/work-like activity.
Interpretive boundary	Lower gamma is a work signal, not direct proof of meaning.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

residualized_lower_gamma

Attempts to remove variance associated with respiration or other available covariates before recomputing lower-gamma contrasts.

Primary outputs	eeg_low_gamma_residualized_contrasts.csv
Why it matters	Separates candidate neural work from common-drive physiology when possible.
Interpretive boundary	Residualization is only as good as the available covariates.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

gammascalpel

Builds task-gamma, meaning-gamma, visual-gamma, and artifact-sensitive gamma scores from spatial/spectral components.

Primary outputs	gammascalpel_window_scores.csv, gammascalpel_score_contrasts.csv
Why it matters	Aims to separate analytic task-lock from candidate

Interpretive boundary	meaning-related work.
Failure/weakness mode	It is a scalp-level heuristic, not an anatomical or cellular measurement. Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

task_lock_plv

Summarizes phase-locking/rigidity-style task engagement features where available.

Primary outputs	phase-lock or contrast tables when available
Why it matters	Helps distinguish forced analytic order from receptive narrative state.
Interpretive boundary	High order is not automatically meaning; it may be task rigidity.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

pac_exploratory

Exploratory theta-gamma phase-amplitude coupling scan.

Primary outputs	PAC tables when enabled
Why it matters	Looks for coupling between slow integration and faster work signals.
Interpretive boundary	PAC is highly artifact-sensitive and should remain exploratory unless preregistered.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

autonomic_api

Computes heart-rate, HRV/RMSSD, and Autonomic Physiology Index-like summaries from cardiac streams when available.

Primary outputs	autonomic_api tables when available
Why it matters	Adds body-state validation to EEG findings.
Interpretive boundary	Cardiac response can reflect arousal, breathing, posture, threat, or emotion.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

respiration_summary

Summarizes respiration belt rate, depth, variability, and condition differences.

Primary outputs	respiration_summary.csv
Why it matters	Controls for respiratory common drive and helps interpret HRV/theta relationships.
Interpretive boundary	Breathing can drive both heart and EEG features; it must not be ignored.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

formal_pncc_theta

Computes formal post-narrative theta carryover contrasts across washout windows.

Primary outputs	pncc_theta_artifact_matched_surrogate_contrasts.csv
Why it matters	Tests whether theta integration persists after a branch ends.
Interpretive boundary	Video-end washout may miss meaning that resolves before the end of the clip.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

tau_coh_theta

Estimates theta coherence half-life or persistence measures when the data support it.

Primary outputs	tau_coh_theta_summary.csv
Why it matters	Quantifies decay/persistence rather than only peak amplitude.
Interpretive boundary	Requires sufficient clean post-event data and stable timing.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

surrogate_theta_carryover

Uses surrogate/randomized comparisons for theta carryover where available.

Primary outputs	surrogate PNCC tables
Why it matters	Protects against seeing ordinary fluctuations as narrative afterglow.
Interpretive boundary	Surrogates do not replace external replication.
Failure/weakness mode	Treat as weak or not evaluable when required streams,

markers, timing, artifact controls, or comparison conditions are missing or contaminated.

post_meaning_cascade

Tests whether expected frequency-order patterns occur after candidate meaning events.

Primary outputs	frequency cascade tables
Why it matters	Looks for gamma/theta/alpha ordering consistent with intake -> integration -> settling.
Interpretive boundary	A cascade pattern is not mechanism proof.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

temporal_semantic_proxy

Computes a posterior-temporal/semantic proxy from posterior temporal gamma relative to visual/task/artifact controls.

Primary outputs	temporal_semantic_proxy_window_scores.csv
Why it matters	Creates a scalp-level proxy for semantic/narrative processing.
Interpretive boundary	It is not STS/MTG source localization.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

postpeak_pncc_theta

Finds gamma/TSP peaks and tests theta carryover after their taper rather than only after the video ends.

Primary outputs	postpeak_peak_and_taper_detection.csv, postpeak_pncc_theta_contrasts.csv
Why it matters	Implements the idea that integration follows the meaningful intake peak, not necessarily the clip boundary.
Interpretive boundary	Peak discovery is exploratory unless the rule is frozen before analysis.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

annotation_locked_theta_carryover

Tests predeclared or supplied annotation windows for theta carryover.

Primary outputs	annotation_locked_theta_carryover_contrasts.csv
Why it matters	Lets the analysis test scene/event anchors directly.
Interpretive boundary	Retrospective subject clock-time estimates are weak

Failure/weakness mode

anchors; media-structural anchors are stronger.

Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

gamma_to_theta_handoff

Computes lagged relationships from gamma/TSP features to later theta features.

Primary outputs

gamma_to_theta_handoff_lag_correlations.csv,
gamma_to_theta_handoff_best_lags.csv

Why it matters

Models gamma intake followed by theta integration.

Interpretive boundary

Lagged correlation is not causation; random-anchor controls are still needed.

Failure/weakness mode

Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

temporal_semantic_proxy_to_theta_handoff

Special case of handoff analysis using TSP as the predictor.

Primary outputs

gamma_to_theta_handoff_best_lags.csv

Why it matters

Tests whether semantic-proxy activity precedes theta carryover.

Interpretive boundary

TSP is a proxy; it cannot prove semantic anatomy.

Failure/weakness mode

Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

thresholded_state_update

Searches for candidate state-update windows using thresholded gamma/TSP/theta/artifact criteria.

Primary outputs

thresholded_state_update_candidate_windows.csv,
thresholded_state_update_summary.csv

Why it matters

Sharpens the scalpel when meaning occurs inside the clip rather than at the end.

Interpretive boundary

Physiology-discovered windows remain exploratory in the same run.

Failure/weakness mode

Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

state_locked_handoff_alignment

Aligns candidate handoff events by state transition rather than movie-clock time.

Primary outputs

state_locked_anchor_summary.csv,

	state_locked_epochs_long.csv, state_locked_model_comparison.csv
Why it matters	Reduces temporal smearing across subjects or runs.
Interpretive boundary	State-locking must be preregistered before it becomes confirmatory.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

human_translation_kht

Estimates phase-level K_{HT} between an EEG gamma/work proxy and a human-translation proxy Y_{HT}.

Primary outputs	human_translation_kht_feature_frame.csv, phasewise_summary, cv tables
Why it matters	Tests rough reciprocal loop structure at the human PR-AYC-G level.
Interpretive boundary	K _{HT} is not K _{OSM} and does not prove OSM.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

candidate_local_kht

New v1.4 module: estimates event-level local K_{HT} around predeclared, annotation, state-locked, or physiology-discovered anchors and tests theta handoff.

Primary outputs	candidate_local_kht_analysis.csv, anchor_manifest, random_reference, interpretation.json
Why it matters	Formalizes the Her 113.9 s style exploratory handoff analysis inside the suite.
Interpretive boundary	K alone is not enough; theta handoff alone is not enough; physiology-discovered anchors remain exploratory.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

media_covariate_inventory

Inventories supplied media QC, stimulus fingerprints, cue schedules, and manifests.

Primary outputs	human_translation_kht_media_covariate_inventory.csv
Why it matters	Keeps raw sensory/input information separate from inferred biology.
Interpretive boundary	Media covariates belong to u(t), not Y(t).
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions

are missing or contaminated.

constrained_dtw_exploratory

Optional dynamic-time-warping alignment for state-transition shapes.

Primary outputs	DTW readiness notes / exploratory tables
Why it matters	Can compare shapes of state transitions despite speed differences.
Interpretive boundary	DTW can overfit; use only as exploratory until locked.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

stimulus_style_windows

Applies stimulus-style presets such as delayed reveal, sustained early peak, threat reveal, or weak/ambiguous.

Primary outputs	configured windows in reports/tables
Why it matters	Makes the suite sensitive to different narrative structures.
Interpretive boundary	Style presets are analysis assumptions, not facts.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

master_endpoint_table

Summarizes major endpoint statuses across modules.

Primary outputs	master_endpoint_status_table.csv
Why it matters	Provides a single executive table for interpretation.
Interpretive boundary	A summary table cannot rescue weak source modules.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

figures

Generates visual figures for key physiological and endpoint summaries.

Primary outputs	figures folder
Why it matters	Supports review and communication.
Interpretive boundary	Figures are explanatory, not proof.
Failure/weakness mode	Treat as weak or not evaluable when required streams, markers, timing, artifact controls, or comparison conditions are missing or contaminated.

CandidateLocal_KHT_v0.1 - formal event rule

CandidateLocal_KHT estimates a local reciprocal coupling between an EEG gamma/work proxy $E(t)$ and a human-translation proxy $Y_{HT}(t)$ around candidate anchors. It then checks whether theta carryover follows the candidate event.

Rung 00 translation rule: K alone is not enough. Theta handoff alone is not enough. A final human-event lock requires both, plus valid anchor status and artifact/timing controls.

$$E_{next} \sim E_t + Y_{HT_t} + artifact_t$$
$$Y_{HT_{next}} \sim Y_{HT_t} + E_t + artifact_t$$
$$\eta = Y_{HT} \rightarrow E \text{ coefficient}$$
$$\zeta = E \rightarrow Y_{HT} \text{ coefficient}$$
$$K_{local} = \sqrt{\text{abs}(\eta * \zeta)}$$

Anchor source hierarchy

Predeclared/media-structural anchor: strongest timing source; can support confirmatory tests if loaded before run or before physiological analysis.

Annotation-file anchor: useful secondary anchor; confirmatory only if preregistered or externally frozen.

State-locked/postpeak anchor: useful algorithmic anchor if the rule is frozen before analysis.

Physiology-discovered anchor: always exploratory within the same run; useful for future freezing only.

Recommended interpretation language

Use: "candidate local K_{HT} support" or "future-freeze anchor candidate." Avoid: "confirmed K_{HT} ," "OSM proof," "Y_OSM detected," or "microtubular mechanism measured."

Output table fields

Field	Meaning
anchor_id	Name of candidate anchor.
condition	CONTROL_1, TARGET_1, or CONTEXTUAL_OVERRIDE_1.
time_sec	Condition-relative candidate time.
K_local	Event-level reciprocal coupling magnitude.
K_local_percentile_vs_random_same_condition	Random-anchor specificity percentile.
delta_theta_0_10_vs_pre	Immediate theta carryover after anchor.
delta_theta_10_30_vs_pre	Delayed theta carryover after anchor.
candidate_local_kht_lock	True only when anchor status and both local K and theta rules pass.